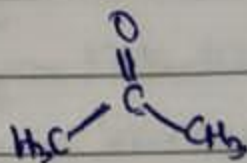


$\nu(C=O)$

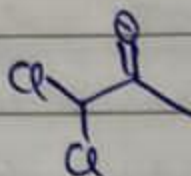


1715 cm⁻¹



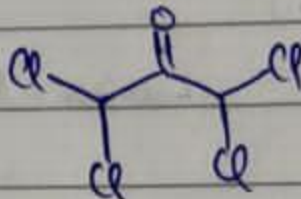
1725 cm⁻¹

(10)



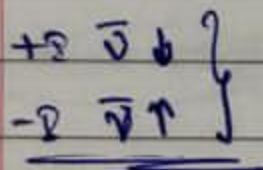
(20)

1740 cm⁻¹



(30)

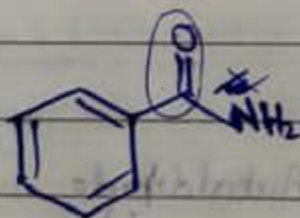
1750 cm⁻¹



Conjugation

Conjugation ↑, $\bar{v}$ ↓

$\bar{v}$ $\nu(C=O)$



Benzamide

1693 cm⁻¹



Methyl benzoate

1730 cm⁻¹

* O more electro-ve, less conjugation, $\bar{v}$ ↑.

Ketones

Aldehydes

Anides

Esters

} comparison

Region: Below 1000 cm^{-1}
 Deformation at $\sim 700 \text{ cm}^{-1}$

cis alkene

Deformation at $\sim 970 - 960 \text{ cm}^{-1}$

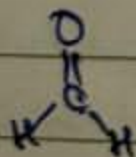
Trans alkene

* $\bar{\nu}$ calculated $\neq \bar{\nu}$ (neighboring atoms, H-bonding, etc.) - 2017

27-2-26 Factors Influencing Vibrational Frequencies

1. Inductive effect
2. Conjugation
3. Hydrogen bonding
4. Bond angle
5. Field effect
6. Sample state

Formaldehyde



1750 cm^{-1}

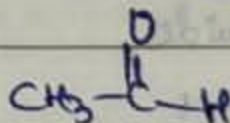
$\nu(\text{C}=\text{O})$

Acetone



1715 cm^{-1}

Acetaldehyde



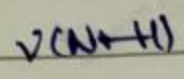
~~1715~~ 1745 cm^{-1}

Inductive effect +I donating
 -I withdrawing

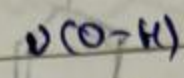
+I - lengthening/weakening of bond
 Bond strength / force constant $\downarrow$

$\bar{\nu} \downarrow$

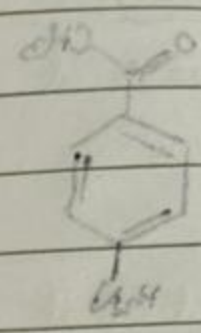
Intermolecular H-bonding



Amines



Alcohols



* O more electronegative than N;

Alcohol H-bonding stronger.

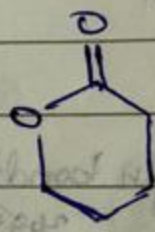
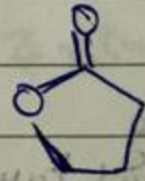
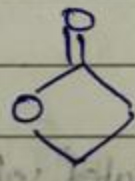
* Decreases more.

3000 cm^{-1}
3300 cm^{-1}

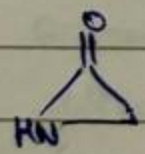
3600 cm^{-1}
3300 cm^{-1} } larger difference

Bond Angle

ketones

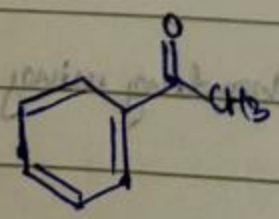


ketone

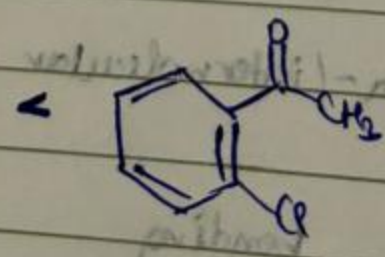


* Ring size $\downarrow$, strain $\uparrow$, force const. $\uparrow$, ν $\uparrow$.

Field Effect



Acetophenone



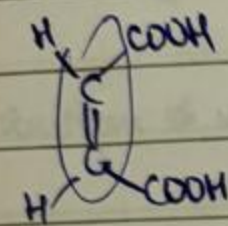
o-chloro

Acetophenone

* If CH_3 in place of Cl , o-chloro is higher.

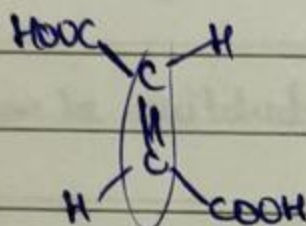
- * Field effects - only o-substituted compounds
- * O and Cl L.P. $\rightarrow$ electrostatic repulsion.
- * $C=O$ bond goes out of plane, lower state of hybridization. Conjugation $\downarrow$, ΔT .
- * Steric hindrance $\uparrow$, ΔT force constant $\uparrow$, ΔT .

Geometric isomers



Maleic acid

cis



Fumaric acid

trans

Factor - steric hindrance

Instrumentation

IR spectroscopy

FTIR

- * IR, FTIR - same, but different instruments

* Single spectrum

* Number of scans

(scan at a time)

Monochromatic light

Polychromatic light

Poor resolution

Better resolution

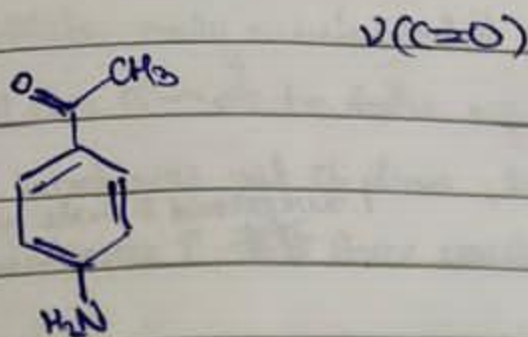
slow technique

fast technique

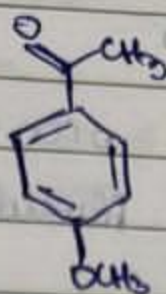
Sample preparation

No sample preparation reqd.; ATR.

(Attenuated total reflectance)

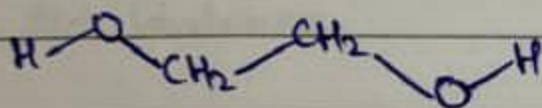


p-Amino
Acetophenone
 1677 cm^{-1}
Conjugation $\uparrow$, $\bar{\nu}$ $\downarrow$



p-Methoxy
Acetophenone
 1684 cm^{-1}

$\nu(\text{O-H})$



Ethylene glycol
Intramolecular hydrogen bonding
 3644 cm^{-1} ("free" value) 3495 cm^{-1} ("associated" value)
(Intra-)

- * IR can distinguish between intra-, intermolecular H-bonding.
- * Intra-peaks sharp and well-defined
- Inter-broad peaks

- * H-bonding $\uparrow$, $\bar{\nu}$ $\downarrow$
- Intra-decrease less
- Inter-decrease more

Q) How do you distinguish between intra- / intermolecular H-bonding using IR spectroscopy?

Intramolecular hydrogen bonding
 3644 cm^{-1} 3495 cm^{-1}

Constructive

OPD = 0, λ , 2λ , 3λ , etc.

Integral no. of wavelengths.

Destructive

OPD = $\frac{\lambda}{2}$, $\frac{3\lambda}{2}$, $\frac{5\lambda}{2}$, etc.

Half integral no. of wavelengths.

* Positions of mirrors can be switched.

1. Nernst Glower - radiation source for FTIR spectrophotometer

High Temp 1250°C upto 7100 cm^{-1} (max radiation)

Sintered mixture of rare earth oxides

2. Globar - Rod of silicon carbide

T $1300 - 1700^\circ\text{C}$

Max. radiation at 820 cm^{-1}

3. Tungsten Filament lamp

(Near IR)

4. Mercury arc lamp

(Far IR)

Dispersive Infrared Spectrophotometer

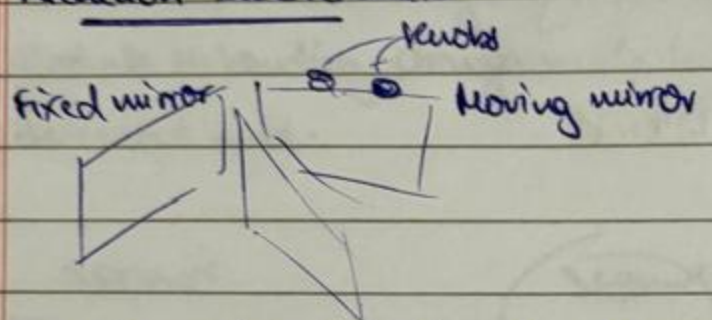
Diagram - Pavia

olive eggs

Fourier Transform Infrared Spectrophotometer

Diagram - Pavia

* Michelson Interferometer



* Interferometric infrared spectroscopy

* Output - interferogram

CP - previously frequency domain spectrum)

* Time domain spectrum

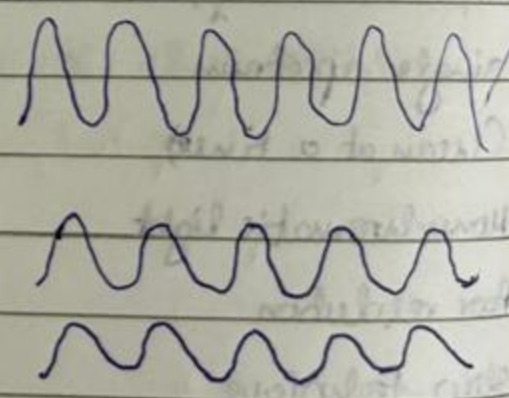
* FT - used to convert time domain spectrum $\rightarrow$ frequency domain spectrum

Graph for analysis

* Mirror adjustment

Constructive and destructive interference

path diff.



Q) Labeled diag. of FTIR spectrophotometer, 2A/OPA

Zero Path Difference (ZPD)

Two arms of the interferometer are equal optical path length.

Q) Differentiate draw diagram

(Constructive)

would must be in same phase

Optical Path Difference (OPD)

Unequal optical path length.

(Destructive)

Diagram